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Title: Clinical guideline for management of Severe Combined Immunodeficiency (SCID)

Clinical guideline for management of Severe Combined Immunodeficiency SCID

Introduction:

Severe Combined Immunodeficiency (SCID) is a heterogeneous complex group of disorders that is related to a monogenic change – found in up to 90% of cases – which ultimately leads to a clinically severe inborn error of immunity. The hallmark of this disease is impairment in T lymphocyte development, inevitably B lymphocyte dysfunction and often a defective innate immunity. Due to the high incidence of severe recurrent or persistent infections, SCID is typically fatal in the first two years of life without a definitive treatment.

Epidemiology of SCID:

Epidemiological studies from databases worldwide show geographical variation. This is mainly due to the genetic background and the level of consanguinity in the community. In K.S.A, Al-Mousa H et al has reported the incidence of SCID is as high as 1.0/2,906 live births using newborn screening diagnosis as compared to 1.0/58,000 infants in the U.S.A.

In Oman, Al-Tamimi S et al has estimated the prevalence of primary immunodeficiency diseases PID as 7.0/100,000 population. The exact prevalence and incidence of children with SCID is not reported, however 17.8% of reported children with PID has combined immunodeficiency CID. This is in line with the incidence of CID in Saudi Arabia (17%). Hence, we expect a higher incidence of SCID in Oman as compared to western communities.

Obviously, the incidence of SCID is more accurately measured by using newborn screening program as compared to retrospective reporting. Therefore, we are looking forward to define the exact incidence of SCID in Omani population by implementing newborn screening in the near future.

Diagnosis:

In the absence of newborn screening using T cell receptor excision circles (TRECs), and in the presence of clinical pointers and/or positive family history, the following should be considered:

A. Criteria for typical SCID:

- absent or very low numbers of CD3+ T cells <300/uL
- absent or very low T cell function (<10% lower range of normal), as measured by proliferation to mitogens

For the diagnosis of atypical SCID and Omenn syndrome, please discuss with pediatric immunologist.

B. Supplementary and additional testing (if available):

- Full blood count.
- Serum immunoglobulins.
- Lymphocyte subset enumeration.
- Detailed T and B cell flowcytometry (including naïve cell, recent thymic emigrants and activated cells).
- V-beta repertoire.
- Microarray.
- Measurement of adenosine deaminase ADA/ Purine nucleoside phosphorylase PNP levels.
- Fibroblast culture for radiosensitivity testing.

C. Genetic Testing

- Targeted mutation.
- Targeted panel/next generation sequencing.
- Whole exome sequencing.

Medical/Supportive care:

The main aim for the newly diagnosed SCID is to prevent infections and avoid system organ damage as both have adverse impact on overall outcome. Therefore, the international recommendations suggest a period of 6 to 8 weeks for the diagnosis-to-hematopoietic stem cell transplant HSCT time in order to minimise the inevitable risk in morbidity and mortality.

The initial medical care following diagnosis should include the following:

A. Breastfeeding:

- Breastfeeding for children should be advocated, however the newly diagnosed baby with SCID can contract CMV infection from breastmilk if mum is CMV seropositive during her pregnancy. Therefore, it is recommended to check the CMV status of mum before initiating breastfeeding.
- If breastfeeding is contraindicated, ready-to-feed infant formula and powdered infant milk formula can be used. Attention to use a boiled water for feed preparation.

B. Vaccination:

- When a diagnosis of SCID is suspected or confirmed, all vaccinations (live or inactive) should be avoided and deferred until after curative treatment and normal immune system functions.

C. Infection prophylaxis:

Children with SCID are at high risk of acquiring not only common childhood infections, but also are at risk of having opportunistic infections including bacterial, mycobacterial, fungal, viral and protozoal. To minimise this risk, it is recommended to initiate the following:

- Immunoglobulin replacement therapy (IRT) should be commenced with a starting dose 0.4-0.6g/kg/3-4 weeks. IRT should be commenced even when the initial Immunoglobulin G (IgG) level is normal, as this is due to maternal transference and should not last long.
- Cotrimoxazole prophylaxis from the 4th week of age should be started. There are variable regimens, however a dose of 5mg/kg/day of trimethoprim component three days per week or daily, either can be followed.
- Fluconazole prophylaxis at a dose of 5mg/kg /day (every 72 hours if neonate <2 weeks of age and every 48 hours if neonate 2-4 weeks of age). In the presence of neutropenia, consider itraconazole as an alternative with a dose of 5 mg/kg/day.
- Acyclovir prophylaxis may be considered if any caregiver with direct patient contact has a history of frequent cold sores. Patients <6 months; 300mg/m² three times a day. Over 6 months of age; 100mg three times a day.
- Palivizumab prophylaxis through winter months if available.

D. BCG infection:

- If there are local changes at the site of the BCG injection, BCGitis with regional lymph node involvement or BCGosis: anti-tuberculosis treatment according to local guideline continued till complete immunological reconstitution.

E. Hospitalisation

Many times, children with SCID are admitted in hospital for various reasons during their course of illness. Promptly following diagnosis, SCID patients need to be admitted in a single room as per the local protective isolation guidelines (preferably isolation room with positive pressure system). The following bundle is recommended upon admission:

- Staff members caring for the child must be well and not have active infection.
- Assessment of nutrition and growth is important. Some children will require total parenteral nutrition TPN to gain enough weight that allows them to have HSCT.
- If the child needs blood products, they should be CMV negative, leuko-filtered and irradiated.
- Patients should undergo regular screening for CMV, Epstein Barr Virus (EBV), adenovirus and HHV6/7, if available. Serological testing is unsuitable for patients with SCID, therefore please ask for PCR.
- Utmost effort must be made to identify infections and treat them effectively for enough period to stabilise and move to HSCT within an urgent timeframe.
- Those with system organ dysfunction will need special attention when using antimicrobials and other pharmacological agents. Referring to the concerned subspecialty is often needed to restore or minimise organ dysfunction.
- Limit venepuncture to prevent iatrogenic anaemia and investigations should be performed within the room where possible.
- If radiosensitive SCID is suspected, it is essential to minimize radiation exposure prior to HSCT.
- Urgent donor search for HSCT is required. All suitable sibling(s) should be screened for match by human leukocyte antigen HLA. In some situations matched family donor can be used. However, if no match is found, using a parent as haploidentical donor can be an option.

F. Family care:

Families of newly diagnosed SCID may need social, emotional, financial and/or other forms of support. Please consider these needs and refer to the suitable service as soon as possible.

All families of newly diagnosed SCID should be updated about the seriousness of the disease. Unfortunately, some families already have unpleasant experience with a previous affected member. Therefore, the following points should be discussed:

- SCID is inherited disease. In our community, the most likely cause of inheritance is autosomal recessive. If this is the case, subsequent pregnancies carry a risk of having affected baby (25%) or a carrier (50%).
- All parents would be provided with a clinical letter that can be used as a reference. It contains information about SCID, instructions to avoid vaccination and early access to medical service for the patient and subsequent siblings.
- According to the circumstances, parents can be referred to preimplantation genetic diagnosis PGD clinic to discuss options of future pregnancies.

Curative options:

A. Haematopoietic Stem Cell Transplantation (HSCT):

- HSCT is the most urgent curative treatment option practised worldwide for children with SCID. It should only be undertaken in a specialist centre with suitable expertise and facilities to isolate and manage. High success rate is seen in children with fully matched donor reaching to upto 90 %, with earlier age of transplant and with no prior active infections.
- PEG-ADA; Enzyme replacement therapy (ERT) is available for adenosine deaminase deficiency (ADA) SCID. If available, can be given for this type of SCID awaiting for HSCT as ERT will only provide short term immune reconstitution.

B. Gene Therapy: approved for X-SCID and ADA-SCID. Clinical trials are ongoing for more types.

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